

On characterization of fuzzy vectors and its applications to fuzzy mathematical programming problems

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Abstract

In this paper, we consider fuzzy mathematical programming problems which involve fuzzy vectors only in the objective function. First, we define fuzzy vectors on n -dimensional Euclidean space $\mathbb{R}^n$ and investigate their properties. Especially, we extend the concepts of the fuzzy max order and the possibility and the necessity measures given in fuzzy number to those of fuzzy vectors. Next, based on the results, we consider fuzzy mathematical programming problems, including fuzzy multicriteria optimization problems, which involve fuzzy vectors only in the objective function and we give the concepts of optimal solutions and characterize their properties.

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1. Introduction

Since the seminal paper [20] given by Zadeh, fuzzy sets theory has been applied to various fields of decision making theory including economics, management science and engineering widely. Especially, much attention has been focused on fuzzy linear programming problems, abbreviated by FLP, which involve fuzzy numbers in coefficients of the objective and constraint functions from the theoretical as well as from the computational point of view ([2,12,14,17]). In FLP, since the objective function is fuzzy-valued and there is no universal concepts of optimal solutions to be accepted widely, it is important task to define the concepts of optimal solutions and investigate their properties. Tanaka et al. [17] formulated FLP as parametric linear programming problems in order to obtain a reasonable optimal solution to FLP. While Luhandjura [12] formulated FLP as semi-infinite linear programming problems with infinite many objective functions. Maeda [14] considered FLP which involve fuzzy numbers only in coefficients of objective functions and defined three types of optimal solutions and investigated the properties of the optimal solutions and the optimal values. However, in those researches, interactivity among fuzzy numbers has not been considered and the assumption that the fuzzy numbers are non-interactive has played an important role. The properties of interaction among fuzzy numbers have been considered [3,7–9,18,19]. Dubois and Prade [3] showed that linear combinations of interactive LR type fuzzy

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numbers could be expressed by simple forms. Tanaka and Ishibuchi [18] investigated the properties of possibilistic linear systems of interactive fuzzy numbers with quadratic membership functions.

While in fuzzy mathematical programming problems, abbreviated by FMP, with interactive fuzzy numbers, Inuiguchi and Sakawa [8] considered a FLP problem with quadratic membership function and showed equivalent conditions between a possibilistic linear programming problem with a quadratic membership function and a stochastic linear programming problem with a multivariate normal distribution. Recently, Inuiguchi et al. [7] defined a new type of interactive fuzzy numbers, called the oblique fuzzy vector, which is defined by a linear transformation of non-interactive fuzzy numbers and investigated their properties. In addition, using possibility approaches, they showed that FLP with oblique fuzzy vectors can be reduced to a linear programming problem and gave the numerical method to solve FLP with oblique fuzzy vectors.

This paper is a continuation of our previous paper [14]. In this paper, we consider FMP, including fuzzy multicriteria optimization problems, which involve fuzzy vectors only in the objective function and define the concepts of optimal solutions and characterize optimal solutions and optimal values to FMP. For that purpose, first, we investigate the properties of fuzzy vectors defined on $\mathbb{R}^n$ and extend the concepts of fuzzy max order given in fuzzy numbers to those of fuzzy vectors. Next, using the ordering, we define the three types of optimal solutions and show that FMP can be reduced to set-valued optimization problems.

This paper is organized as follows. In Section 2, we give some definitions and notations on fuzzy numbers. In Section 3, we investigate basic properties of fuzzy vectors. Especially, we extend the concepts of fuzzy max order, possibility measure and necessity measure which are defined on fuzzy number to those of fuzzy vectors. In Section 4, we consider a FMP, and define the concepts of optimal solutions and characterize their optimal solutions and give some numerical examples.

2. Preliminary

In this section, we shall give some definitions and notations on fuzzy numbers, which are used throughout the paper.

Let $\mathbb{R}^n$ be n -dimensional Euclidean space. We denote the non-negative orthant and non-positive orthant by $\mathbb{R}_+^n$ and $\mathbb{R}_-^n$, respectively. For any two vectors $x, y \in \mathbb{R}^n$, we write $x \geq y$ iff $x - y \in \mathbb{R}_+^n$, $x \gg y$ iff $x - y \in \mathbb{R}_+^n$ and $x \neq y$ and $x > y$ iff $x - y \in \text{int } \mathbb{R}_+^n$, respectively, where int denotes the interior of sets. Let $x \equiv (x_1, x_2, \dots, x_n)^T$, $y \equiv (y_1, y_2, \dots, y_n)^T \in \mathbb{R}^n$ be any two vectors, where $x_i, y_i \in \mathbb{R}$, $i = 1, 2, \dots, n$ and T denotes the transpose of the vector. Then we denote the inner product of x and y by $\langle x, y \rangle$.

Let $A \subseteq \mathbb{R}^n$ be any non-empty set. Then a point $a \in A$ is a maximal element in A iff there is no $x \in A$ such that $a \leq x$ holds. While a point a is a minimal element in A iff there is no $x \in A$ such that $x \leq a$ holds. We denote the sets of all maximal elements and all minimal elements in A by $\sup A$ and $\inf A$, respectively. Let $A, B \subseteq \mathbb{R}^n$ be any non-empty sets. We write $A + B \equiv \{z \in \mathbb{R}^n \mid z = x + y, x \in A, y \in B\}$. Let $A \subseteq \mathbb{R}^n$ be any non-empty convex set and let $x \in A$ be any element. Then the set $N(x; A) \equiv \{d \in \mathbb{R}^n \mid \langle d, y - x \rangle \leq 0, \forall y \in A\}$ is said to be a normal cone to A at x . By definition, the normal cone $N(x; A)$ is closed and convex.

Definition 2.1. A fuzzy vector $\tilde{a}$ on $\mathbb{R}^n$ is defined as a fuzzy set on n -dimensional Euclidean space $\mathbb{R}^n$, whose membership function $\mu_{\tilde{a}}$ satisfies the following conditions:

- (i) $\mu_{\tilde{a}}$ is a mapping from $\mathbb{R}^n$ to the closed interval $[0, 1]$,
- (ii) there exists a unique vector $c \in \mathbb{R}^n$, called center of $\tilde{a}$, such that $\mu_{\tilde{a}}(c) = 1$,
- (iii) $\mu_{\tilde{a}}$ is upper semi-continuous on $\mathbb{R}^n$,
- (iv) $\mu_{\tilde{a}}$ is quasi-concave on $\mathbb{R}^n$.

For any fuzzy vector $\tilde{a}$, the set $\text{supp}(\tilde{a}) \equiv \text{cl}\{x \in \mathbb{R}^n \mid \mu_{\tilde{a}}(x) > 0\}$ is called the support of $\tilde{a}$, where cl denotes the closure of the set. By $\mathcal{F}(\mathbb{R}^n)$, we denote the set of all fuzzy vectors whose supports are bounded.

In Definition 2.1, when $n = 1$, a fuzzy vector $\tilde{a} \in \mathcal{F}(\mathbb{R}^1)$ is called fuzzy number and we denote the set of all fuzzy numbers by $\mathcal{F}(\mathbb{R})$.

Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector and let $\alpha \in (0, 1]$ be any real number. Then the set $\tilde{a}_\alpha \equiv \{x \in \mathbb{R}^n \mid \mu_{\tilde{a}}(x) \geq \alpha\}$ is called the α -level set of $\tilde{a}$. For $\alpha = 0$, we set $\tilde{a}_0 \equiv \text{supp}(\tilde{a})$. The fuzzy vector $\tilde{a}$ is said to be robust if $\text{cl}\{x \in \mathbb{R}^n \mid \mu_{\tilde{a}}(x) > \alpha\} = \tilde{a}_\alpha$ holds for all $\alpha \in [0, 1)$ (see [5]). If the α -level set $\tilde{a}_\alpha$ is rectangle in $\mathbb{R}^n$, the fuzzy vector $\tilde{a}$ is called

rectangle type fuzzy vector, and we denote the set of all rectangle type fuzzy vectors by $\mathcal{F}_r(\mathbb{R}^n)$. It is obvious that $\mathcal{F}(\mathbb{R}) = \mathcal{F}_r(\mathbb{R})$ holds.

The following theorem shows that any fuzzy numbers, more generally fuzzy sets, are characterized by the family of the α level sets.

Theorem 1 (Representation theorem [11]). *Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector. Then the following holds.*

$$\mu_{\tilde{a}}(x) = \sup_{\alpha} \min\{\alpha, I_{\tilde{a}_\alpha}(x)\},$$

where $I_{\tilde{a}_\alpha}(\cdot)$ is the indicator function.

Definition 2.2. Let L be a function from $\mathbb{R}$ to $\mathbb{R}$ satisfying the following conditions:

- (i) $L(x) = L(-x) \quad \forall x \in \mathbb{R}$,
- (ii) $L(x) = 1$ iff $x = 0$,
- (iii) $L(\cdot)$ is upper semi-continuous on $\mathbb{R}$,
- (iv) $L(\cdot)$ is non-increasing on $[0, +\infty)$ and there exists a positive number $x_0 \in (0, +\infty)$ such that $x_0 = \inf\{x > 0 \mid L(x) = 0\}$.

Then the function L is called a shape function, and we call the point x_0 in (iv) the zero point of L .

Definition 2.3. Let m be any real number and let h, k be any positive numbers. Let L and R be any shape functions. A fuzzy number $\tilde{a}$ whose membership function is given by the following formula:

$$\mu_{\tilde{a}}(x) \equiv \begin{cases} L\left(\frac{m-x}{h}\right) & \text{for } x \leq m, \\ R\left(\frac{x-m}{k}\right) & \text{for } m \leq x \end{cases} \quad (1)$$

is called an LR fuzzy number, and we denote the set of all LR fuzzy numbers by $\mathcal{F}_{LR}(\mathbb{R})$.

By the definition, the real number m in (1) is equal to the center of $\tilde{a}$, and we call the real numbers h, k the deviation parameters of $\tilde{a}$. Since any LR fuzzy number $\tilde{a}$ is characterized by the center m and the deviation parameters h, k of $\tilde{a}$, we denote the LR fuzzy number $\tilde{a}$ by $\tilde{a} \equiv (m, h, k)_{LR}$. In particular, LR fuzzy number is said to be symmetric if $h = k$ and $L(x) = R(x)$ for all $x \in \mathbb{R}$, and we denote the symmetric LR fuzzy number $\tilde{a}$ by $\tilde{a} \equiv (m, h)_L$.

Let $\tilde{a} \equiv (a, h, k)_{LR} \in \mathcal{F}_{LR}(\mathbb{R})$ be any LR fuzzy number and let $\alpha \in [0, 1]$ be any real number. Then, we have $a - t_\alpha^L h = \inf \tilde{a}_\alpha$ and $a + t_\alpha^R k = \sup \tilde{a}_\alpha$, where $t_\alpha^L \equiv \max\{t \in \mathbb{R} \mid L(t) \geq \alpha\}$ and $t_\alpha^R \equiv \max\{t \in \mathbb{R} \mid R(t) \geq \alpha\}$. By setting $\tilde{a}_\alpha^L \equiv \inf \tilde{a}_\alpha$ and $\tilde{a}_\alpha^R \equiv \sup \tilde{a}_\alpha$, then it holds that $\tilde{a}_\alpha = [\tilde{a}_\alpha^L, \tilde{a}_\alpha^R]$.

3. Properties of fuzzy vectors

In this section, we shall investigate the properties of fuzzy vectors and extend some definitions which are well known in fuzzy numbers to those of fuzzy vectors.

Let $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vectors and let $\lambda \in \mathbb{R}$ be any scalar. Then the sum $\tilde{a} + \tilde{b}$, the scalar product $\lambda \tilde{a}$ and the inner product $\langle \tilde{a}, \tilde{b} \rangle$ are defined by membership functions

$$\mu_{\tilde{a}+\tilde{b}}(t) = \sup_{t=u+v} \min\{\mu_{\tilde{a}}(u), \mu_{\tilde{b}}(v)\}, \quad (2)$$

$$\mu_{\lambda \tilde{a}}(t) = \max \left\{ 0, \sup_{t=\lambda u} \mu_{\tilde{a}}(u) \right\}, \quad (3)$$

$$\mu_{\langle \tilde{a}, \tilde{b} \rangle}(t) = \sup_{t=\langle u, v \rangle} \min\{\mu_{\tilde{a}}(u), \mu_{\tilde{b}}(v)\}, \quad (4)$$

where we set $\sup\{\emptyset\} = -\infty$. Moreover, for any fuzzy vector $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ and any vector $b \in \mathbb{R}^n$, the inner product $\langle \tilde{a}, b \rangle$ which is defined by

$$\begin{aligned}\mu_{\langle \tilde{a}, b \rangle}(t) &= \sup_{t=\langle u, b \rangle} \mu_{\tilde{a}}(u), \\ &= \sup_{\alpha \in [0, 1]} \min\{\alpha, I_{\langle \tilde{a}, b \rangle}(t)\} \quad \forall t \in \mathbb{R}\end{aligned}$$

is a fuzzy number [10], where $\langle \tilde{a}_\alpha, b \rangle \equiv \{t \in \mathbb{R} \mid t = \langle u, b \rangle, u \in \tilde{a}_\alpha\}$. Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector and we set $\tilde{a}_i \equiv \langle \tilde{a}, e^i \rangle$, $i = 1, 2, \dots, n$, where e^i is the unit vector whose i the element is one and other elements are all zeros. Then we define the n -dimensional fuzzy vector $(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)$ by

$$\mu_{(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)}(x_1, x_2, \dots, x_n) \equiv \min_{i=1, 2, \dots, n} \mu_{\tilde{a}_i}(x_i).$$

By definition, for each $(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)$, it holds that

$$(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)_\alpha = \tilde{a}_{1\alpha} \times \tilde{a}_{2\alpha} \times \dots \times \tilde{a}_{n\alpha} \quad \forall \alpha \in [0, 1].$$

Namely, $(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n) \in \mathcal{F}_r(\mathbb{R}^n)$.

Then we have the following results.

Theorem 2. Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector. Then we have

$$\tilde{a}_\alpha \subseteq (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)_\alpha \quad \forall \alpha \in [0, 1]. \quad (5)$$

While, if $\tilde{a} \in \mathcal{F}_r(\mathbb{R}^n)$, we have

$$(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)_\alpha = \tilde{a}_\alpha \quad \forall \alpha \in [0, 1]. \quad (6)$$

Proof. For any $x \equiv (x_1, x_2, \dots, x_n) \in \tilde{a}_\alpha$,

$$\begin{aligned}\mu_{\tilde{a}_i}(x_i) &\equiv \sup\{\mu_{\tilde{a}}(u) \mid \langle u, e^i \rangle = x_i\} \\ &\geq \mu_{\tilde{a}}(x) \geq \alpha, \quad i = 1, 2, \dots, n.\end{aligned}$$

Therefore, we have $\mu_{(\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)}(x) \geq \alpha$.

Conversely, let $x \equiv (x_1, x_2, \dots, x_n) \in (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)_\alpha$ be any element. Then, it holds that

$$\mu_{\tilde{a}_i}(x_i) = \sup_{\beta \in [0, 1]} \min\{\beta, 1_{\langle \tilde{a}_\alpha, e^i \rangle}(x_i)\} \geq \alpha, \quad i = 1, 2, \dots, n,$$

which implies that $x_i \in \langle \tilde{a}_\alpha, e^i \rangle$. Since $\tilde{a}_\alpha$ is rectangle, $\tilde{a}_\alpha = \prod_{i=1}^n \langle \tilde{a}_\alpha, e^i \rangle$. Hence, we have $(x_1, x_2, \dots, x_n) \in \tilde{a}_\alpha$. $\square$

Example 3.1. Let $\tilde{a} \equiv \widetilde{(2, 3)} \in \mathcal{F}(\mathbb{R}^2)$ be the fuzzy vector defined by (Figs. 1 and 2)

$$\mu_{\tilde{a}}(r_1, r_2) \equiv \max\{1 - (r_1 - 2)^2 - (r_2 - 3)^2, 0\}.$$

Then we have for any $\alpha \in [0, 1]$

$$\tilde{a}_\alpha = \{(r_1, r_2) \in \mathbb{R}^2 \mid (r_1 - 2)^2 + (r_2 - 3)^2 \leq 1 - \alpha\}.$$

On the other hand,

$$(\tilde{a}_1, \tilde{a}_2)_\alpha = [2 - \sqrt{1 - \alpha}, 2 + \sqrt{1 - \alpha}] \times [3 - \sqrt{1 - \alpha}, 3 + \sqrt{1 - \alpha}].$$

Hence we have $\tilde{a}_\alpha \subsetneq (\tilde{a}_1, \tilde{a}_2)_\alpha$, $\alpha \in [0, 1]$.

Next we consider the fuzzy vector $\tilde{b} \equiv \widetilde{(2, 3)} \in \mathcal{F}_r(\mathbb{R}^2)$ defined by (Figs. 3 and 4)

$$\mu_{\tilde{b}}(r_1, r_2) \equiv \max\{\min\{1 - |r_1 - 2|, 1 - |r_2 - 3|\}, 0\}.$$

It is easy to check that $\tilde{b}_\alpha = (\tilde{b}_1, \tilde{b}_2)_\alpha$ for each $\alpha \in [0, 1]$.

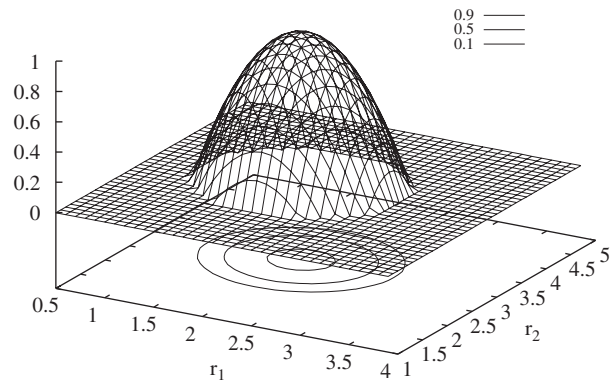


Fig. 1. Fuzzy vector $\tilde{a}$.

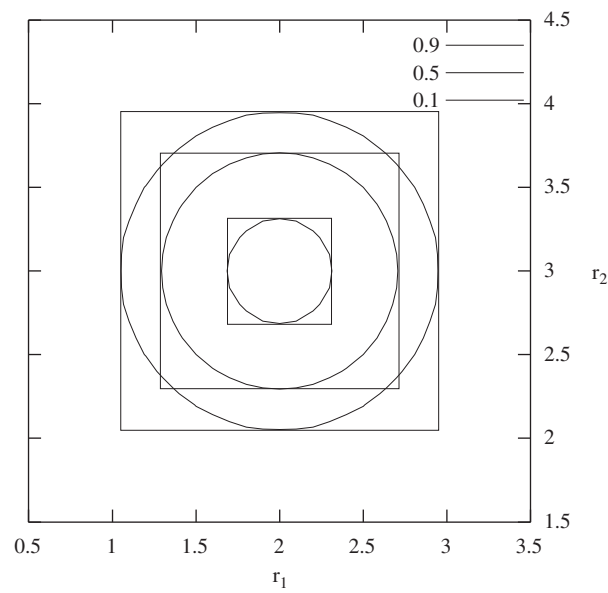


Fig. 2. Level curves of $\tilde{a}$ and $(\tilde{a}_1, \tilde{a}_2)$.

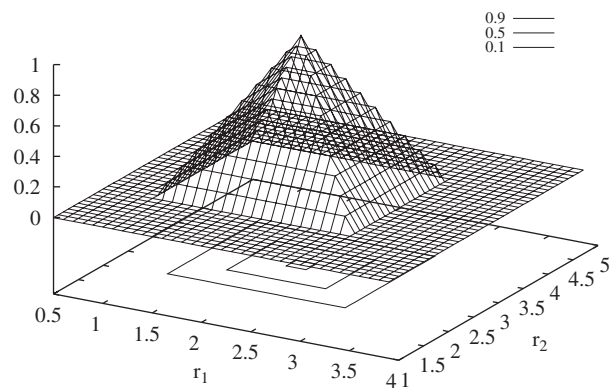
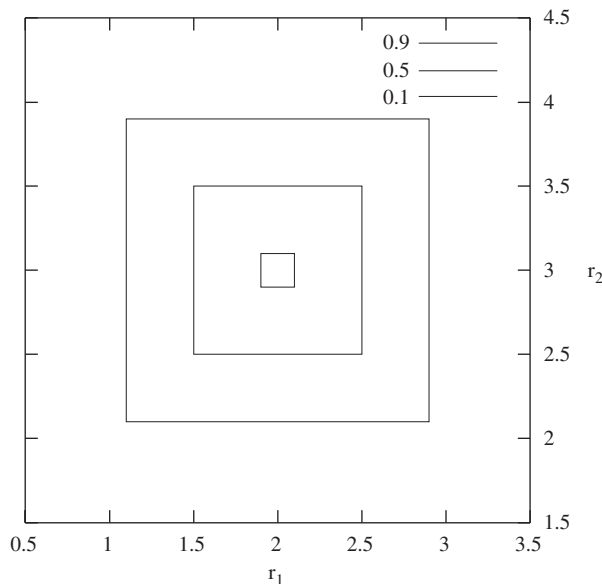


Fig. 3. Fuzzy vector $\tilde{b}$.

Fig. 4. Level curves of $\tilde{b}$ and $(\tilde{b}_1, \tilde{b}_2)$.

Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector and let $\alpha \in [0, 1]$ be any real number. Since α -level set $\tilde{a}_\alpha$ is compact and convex, the $\sup \tilde{a}_\alpha$ and $\inf \tilde{a}_\alpha$ are non-empty, and it holds that

$$\tilde{a}_\alpha \subseteq \inf \tilde{a}_\alpha + \mathbb{R}_+^n \quad \text{and} \quad \tilde{a}_\alpha \subseteq \sup \tilde{a}_\alpha + \mathbb{R}_-^n.$$

We call the sets $\inf \tilde{a}_\alpha + \mathbb{R}_+^n$ and $\sup \tilde{a}_\alpha + \mathbb{R}_-^n$ the maximal upper and the minimum lower half intervals including $\tilde{a}_\alpha$, respectively. By using these properties, we define the following binary relations on $\mathcal{F}(\mathbb{R}^n)$.

Definition 3.1. For any fuzzy vectors $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R}^n)$, we write

$$\tilde{a} \preceq \tilde{b} \text{ iff } \inf \tilde{b}_\alpha \subseteq \inf \tilde{a}_\alpha + \mathbb{R}_+^n \quad \text{and} \quad \sup \tilde{a}_\alpha \subseteq \sup \tilde{b}_\alpha + \mathbb{R}_-^n \quad \forall \alpha \in [0, 1], \quad (7)$$

$$\tilde{a} < \tilde{b} \text{ iff } \inf \tilde{b}_\alpha \subseteq \inf \tilde{a}_\alpha + \text{int } \mathbb{R}_+^n \quad \text{and} \quad \sup \tilde{a}_\alpha \subseteq \sup \tilde{b}_\alpha + \text{int } \mathbb{R}_-^n \quad \forall \alpha \in [0, 1], \quad (8)$$

$$\tilde{a} = \tilde{b} \text{ iff } \tilde{a} \preceq \tilde{b} \quad \text{and} \quad \tilde{b} \preceq \tilde{a}. \quad (9)$$

It is not difficult to see that the binary relation $\preceq$ is a pseudo-ordering on $\mathcal{F}(\mathbb{R}^n)$, and we call it a fuzzy max order.

For simplicity, in Definition 3.1, for each $\alpha \in [0, 1]$, we write (Fig. 5)

$$\tilde{a}_\alpha \preceq_S \tilde{b}_\alpha \text{ iff } \inf \tilde{b}_\alpha \subseteq \inf \tilde{a}_\alpha + \mathbb{R}_+^n \quad \text{and} \quad \sup \tilde{a}_\alpha \subseteq \sup \tilde{b}_\alpha + \mathbb{R}_-^n,$$

$$\tilde{a}_\alpha <_S \tilde{b}_\alpha \text{ iff } \inf \tilde{b}_\alpha \subseteq \inf \tilde{a}_\alpha + \text{int } \mathbb{R}_+^n \quad \text{and} \quad \sup \tilde{a}_\alpha \subseteq \sup \tilde{b}_\alpha + \text{int } \mathbb{R}_-^n,$$

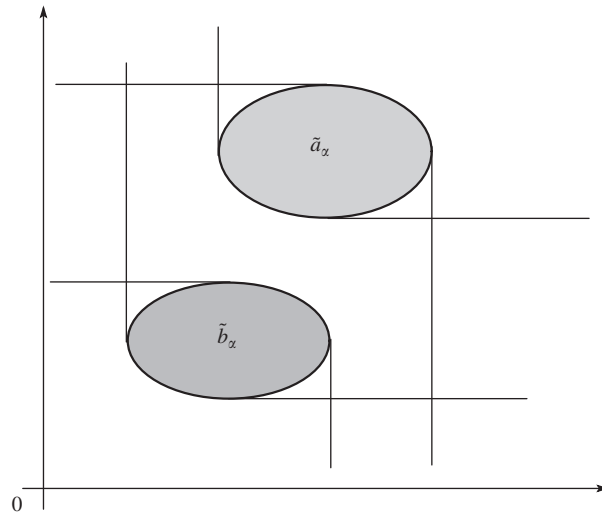
$$\tilde{a}_\alpha =_S \tilde{b}_\alpha \text{ iff } \tilde{a}_\alpha \preceq_S \tilde{b}_\alpha \quad \text{and} \quad \tilde{b}_\alpha \preceq_S \tilde{a}_\alpha.$$

Then the binary relation $\preceq_S$ is a pseudo-ordering on the sets of all non-empty compact convex sets. In case that $\tilde{a}_\alpha$ and $\tilde{b}_\alpha$ are singletons, we have $\tilde{a}_\alpha \preceq_S \tilde{b}_\alpha$ if and only if $\tilde{a}_\alpha \preceq \tilde{b}_\alpha$, and $\tilde{a}_\alpha <_S \tilde{b}_\alpha$ if and only if $\tilde{a}_\alpha < \tilde{b}_\alpha$.

Let $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vectors such that $\tilde{a} \preceq \tilde{b}$ and let $\alpha \in [0, 1]$ be any real number. Then, for each $b \in \inf \tilde{b}_\alpha$, there exists a vector $a \in \inf \tilde{a}_\alpha$ such that $a \preceq b$ holds. On the other hand, for each $a \in \sup \tilde{a}_\alpha$, there exists a vector $b \in \sup \tilde{b}_\alpha$ such that $a \preceq b$ holds. For any fuzzy vectors $\tilde{a}, \tilde{b} \in \mathcal{F}_r(\mathbb{R}^n)$, since the $\inf \tilde{a}_\alpha$, $\inf \tilde{b}_\alpha$, $\sup \tilde{a}_\alpha$ and $\sup \tilde{b}_\alpha$ are singletons, we have

$$\tilde{a} \preceq \tilde{b} \text{ iff } \inf \tilde{a}_\alpha \preceq \inf \tilde{b}_\alpha \quad \text{and} \quad \sup \tilde{a}_\alpha \preceq \sup \tilde{b}_\alpha \quad \forall \alpha \in [0, 1], \quad (10)$$

$$\tilde{a} < \tilde{b} \text{ iff } \inf \tilde{a}_\alpha < \inf \tilde{b}_\alpha \quad \text{and} \quad \sup \tilde{a}_\alpha < \sup \tilde{b}_\alpha \quad \forall \alpha \in [0, 1]. \quad (11)$$

Fig. 5. An example of $\tilde{a}_\alpha \geq_S \tilde{b}_\alpha$.

It is easy to show that the fuzzy max order $\geq$ is a partial ordering on $\mathcal{F}_r(\mathbb{R}^n)$. In particular, since $\mathcal{F}(\mathbb{R}) = \mathcal{F}_r(\mathbb{R})$, the above definition is equal to the definitions given by Ramík [15].

From Theorem 2, for any $\tilde{a} \in \mathcal{F}_r(\mathbb{R}^n)$, we have $\tilde{a} = (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_n)$. Therefore, the following theorem holds.

Theorem 3. For any $\tilde{a}, \tilde{b} \in \mathcal{F}_r(\mathbb{R}^n)$, it holds that

$$\tilde{a} \leq \tilde{b} \text{ iff } \langle \tilde{a}, e^i \rangle \leq \langle \tilde{b}, e^i \rangle, \quad i = 1, 2, \dots, n. \quad (12)$$

Proof. For each $\alpha \in [0, 1]$, from Theorem 2, note that $\tilde{a}_\alpha = (\langle \tilde{a}, e^1 \rangle, \langle \tilde{a}, e^2 \rangle, \dots, \langle \tilde{a}, e^n \rangle)_\alpha$. By Definition 3.1, for each $\alpha \in [0, 1]$, we have

$$\begin{aligned} \tilde{a} \leq \tilde{b} &\text{ iff } \inf \tilde{a}_\alpha \leq \inf \tilde{b}_\alpha \text{ and } \sup \tilde{a}_\alpha \leq \sup \tilde{b}_\alpha, \\ &\text{ iff } \inf \langle \tilde{a}_\alpha, e^i \rangle \leq \inf \langle \tilde{b}_\alpha, e^i \rangle \text{ and } \sup \langle \tilde{a}_\alpha, e^i \rangle \leq \sup \langle \tilde{b}_\alpha, e^i \rangle, \quad i = 1, 2, \dots, n, \end{aligned}$$

which implies that $(\langle \tilde{a}, e^1 \rangle, \langle \tilde{a}, e^2 \rangle, \dots, \langle \tilde{a}, e^n \rangle) \leq (\langle \tilde{b}, e^1 \rangle, \langle \tilde{b}, e^2 \rangle, \dots, \langle \tilde{b}, e^n \rangle)$ holds. $\square$

Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector. Then we define fuzzy sets $\tilde{a}^+$ and $\tilde{a}^-$ by

$$\mu_{\tilde{a}^+}(r) \equiv \sup_{r \geq u} \mu_{\tilde{a}}(u), \quad (13)$$

$$\mu_{\tilde{a}^-}(r) \equiv \sup_{r \leq u} \mu_{\tilde{a}}(u). \quad (14)$$

Then we have the following theorem.

Theorem 4. Let $\tilde{a} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vector. Then, for each $\alpha \in [0, 1]$, we have

$$\tilde{a}_\alpha + \mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid \mu_{\tilde{a}^+}(x) \geq \alpha\}, \quad (15)$$

$$\tilde{a}_\alpha + \mathbb{R}_-^n = \{x \in \mathbb{R}^n \mid \mu_{\tilde{a}^-}(x) \geq \alpha\}. \quad (16)$$

Proof. Let $r \in \tilde{a}_\alpha^+$ be any vector. By definition, we have $\sup_{r \geq u} \mu_{\tilde{a}}(u) \geq \alpha$. First, we consider the case that $\sup_{r \geq u} \mu_{\tilde{a}}(u) > \alpha$ holds. Then there exists a vector $u \leq r$ such that $\mu_{\tilde{a}}(u) > \alpha$. Hence $r \in \tilde{a}_\alpha + \mathbb{R}_+^n$. Next we consider the case that $\sup_{r \geq u} \mu_{\tilde{a}}(u) = \alpha$ holds. For each integer $k \geq 1$, there exists a vector $u_k \leq r$ such that $\alpha \leq \mu_{\tilde{a}}(u_k) < \alpha + 1/k$ holds. Since $\tilde{a}_0$ is compact and $u_k \in \tilde{a}_0$, without loss of generality, we could assume that the sequence $\{u_k\}$ converges a vector $\bar{r} \in \tilde{a}_0$ and $\bar{r} \leq r$. Moreover, by upper semi-continuity of $\mu_{\tilde{a}}$, we have $\mu_{\tilde{a}}(\bar{r}) \geq \liminf_{k \rightarrow +\infty} \mu_{\tilde{a}}(u_k) \geq \alpha$, which implies that $\bar{r} \in \tilde{a}_\alpha$. Therefore, we have $r \in \tilde{a}_\alpha + \mathbb{R}_+^n$.

Conversely, let $r \in \tilde{a}_\alpha + \mathbb{R}_+^n$ be any vector. By definition, there exists a vector $u \in \tilde{a}_\alpha$ such that $u \leq r$. Therefore, $\sup_{r \geq u} \mu_{\tilde{a}}(u) \geq \alpha$, which implies that $r \in \{r \in \mathbb{R}^n \mid \mu_{\tilde{a}^+}(x) \geq \alpha\}$.

By a similar way, we could show that $\tilde{a}_\alpha + \mathbb{R}_-^n = \{x \in \mathbb{R}^n \mid \mu_{\tilde{a}^-}(x) \geq \alpha\}$ holds. $\square$

Let $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R}^n)$ be any fuzzy vectors. Then we define the following relations [4].

$$\text{Pos}(\tilde{a} \geq \tilde{b}) \equiv \sup_u \min\{\mu_{\tilde{a}}(u), \mu_{\tilde{b}^+}(u)\} = \sup_{u \geq v} \min\{\mu_{\tilde{a}}(u), \mu_{\tilde{b}}(v)\},$$

$$\text{Nes}(\tilde{a} \geq \tilde{b}) \equiv \inf_u \max\{1 - \mu_{\tilde{a}}(u), \mu_{\tilde{b}^+}(u)\} = \inf_u \sup_{u \geq v} \max\{1 - \mu_{\tilde{a}}(u), \mu_{\tilde{b}}(v)\}.$$

Then we have the following results.

Theorem 5. Let $\tilde{a}, \tilde{b} \in \mathcal{F}(\mathbb{R}^n)$ be any robust fuzzy vectors. Then, for $\forall \alpha \in (0, 1]$, the following relations hold:

$$(i) \text{Pos}(\tilde{a} \geq \tilde{b}) \geq \alpha \text{ if and only if } \sup \tilde{a}_\alpha \cap (\inf \tilde{b}_\alpha + \mathbb{R}_+^n) \neq \emptyset, \quad (17)$$

$$(ii) \text{Nes}(\tilde{a} \geq \tilde{b}) \geq \alpha \text{ if and only if } \inf \tilde{a}_{1-\alpha} \subseteq \inf \tilde{b}_\alpha + \mathbb{R}_+^n. \quad (18)$$

Proof. (i) Suppose that $\text{Pos}(\tilde{a} \geq \tilde{b}) \geq \alpha$. By definition, there exists a vector $u \in \mathbb{R}^n$ such that $\mu_{\tilde{a}}(u) \geq \alpha$ and $\mu_{\tilde{b}^+}(u) \geq \alpha$ hold. Therefore, $u \in \tilde{a}_\alpha$ and $u \in \tilde{b}_\alpha + \mathbb{R}_+^n$ hold. Since $u \in \tilde{a}_\alpha \subseteq \sup \tilde{a}_\alpha + \mathbb{R}_-^n$ and $u \in \inf \tilde{b}_\alpha + \mathbb{R}_+^n$, we have $\sup \tilde{a}_\alpha \cap (\inf \tilde{b}_\alpha + \mathbb{R}_+^n) \neq \emptyset$.

Conversely, let $u \in \sup \tilde{a}_\alpha \cap (\inf \tilde{b}_\alpha + \mathbb{R}_+^n)$ be any element. Then $\mu_{\tilde{a}}(u) \geq \alpha$ and $\mu_{\tilde{b}^+}(u) \geq \alpha$ hold. Therefore, we have $\text{Pos}(\tilde{a} \geq \tilde{b}) \geq \alpha$.

(ii) Suppose that $\text{Nes}(\tilde{a} \geq \tilde{b}) \geq \alpha$. Then, for each $u \in \mathbb{R}^n$, $1 - \mu_{\tilde{a}}(u) \geq \alpha$, or $\mu_{\tilde{b}^+}(u) \geq \alpha$ holds. Namely, we have $\mathbb{R}^n = \{u \in \mathbb{R}^n \mid \mu_{\tilde{a}}(u) \leq 1 - \alpha\} \cup \{u \in \mathbb{R}^n \mid \mu_{\tilde{b}^+}(u) \geq \alpha\}$. By robustness of $\tilde{a}$, we have $\text{cl}\{u \in \mathbb{R}^n \mid \mu_{\tilde{a}}(u) > 1 - \alpha\} = \{u \in \mathbb{R}^n \mid \mu_{\tilde{a}}(u) \geq 1 - \alpha\} \subseteq \{u \in \mathbb{R}^n \mid \mu_{\tilde{b}^+}(u) \geq \alpha\}$, which implies that $\inf \tilde{a}_{1-\alpha} \subseteq \inf \tilde{b}_\alpha + \mathbb{R}_+^n$.

Conversely, let $\alpha \in [0, 1]$ be any real number and suppose that $\inf \tilde{a}_{1-\alpha} \subseteq \inf \tilde{b}_\alpha + \mathbb{R}_+^n$ holds. Noting that $\tilde{a}_{1-\alpha} \subseteq \tilde{b}_\alpha + \mathbb{R}_+^n$, we have $\mathbb{R}^n = \{u \in \mathbb{R}^n \mid \mu_{\tilde{a}}(u) \leq 1 - \alpha\} \cup \{u \in \mathbb{R}^n \mid \mu_{\tilde{b}^+}(u) \geq \alpha\}$. Hence, each $u \in \mathbb{R}^n$, we have $\max\{1 - \mu_{\tilde{a}}(u), \mu_{\tilde{b}^+}(u)\} \geq \alpha$. Therefore, we have $\text{Nes}(\tilde{a} \geq \tilde{b}) \geq \alpha$. $\square$

For $\tilde{a}, \tilde{b} \in \mathcal{F}_r(\mathbb{R}^n)$, since $\sup \tilde{a}_\alpha$, $\inf \tilde{a}_{1-\alpha}$ and $\inf \tilde{b}_\alpha$ are singletons, we have the following corollary.

Corollary 3.1. Let $\tilde{a}, \tilde{b} \in \mathcal{F}_r(\mathbb{R}^n)$ be any robust fuzzy vectors. Then, for $\forall \alpha \in (0, 1]$, the following relations hold:

$$(i) \text{Pos}(\tilde{a} \geq \tilde{b}) \geq \alpha \text{ if and only if } \sup \tilde{a}_\alpha \geq \inf \tilde{b}_\alpha, \quad (19)$$

$$(ii) \text{Nes}(\tilde{a} \geq \tilde{b}) \geq \alpha \text{ if and only if } \inf \tilde{a}_{1-\alpha} \geq \inf \tilde{b}_\alpha. \quad (20)$$

Corollary 3.1 is well-known results in $\mathcal{F}(\mathbb{R})$, and these relations were proposed by Dubois and Prade [4]. Hence our definitions are extensions of those given for fuzzy numbers.

4. Fuzzy Mathematical Programming Problems

In the previous section, we defined fuzzy max order to fuzzy vectors and investigated its properties. In this section, we consider optimization problems with fuzzy vector-valued objective function and define the concepts of optimal solutions to the problem and investigate their properties.

Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^\ell)$ be any fuzzy vector-valued function and we consider the following problem with fuzzy vector-valued objective function:

$$(\text{FMP}) \quad \begin{cases} \text{maximize} & \tilde{F}(x) \\ \text{subject to} & Ax = b, \ x \geq 0, \end{cases}$$

where $A \in \mathbb{R}^{n \times m}$ and $b \in \mathbb{R}^m$. For simplicity, we set $X = \{x \in \mathbb{R}^n \mid Ax = b, \ x \geq 0\}$.

Problem (FMP) is called a fuzzy multicriteria optimization problem and many authors have proposed the concepts of optimal solutions and investigated their properties ([6,16]).

In this section, based on the fuzzy max order given in Definition 3.1, we define three types of optimal solutions to Problem (FMP).

Definition 4.1. A point $x^* \in X$ is said to be an optimal solution to Problem (FMP) iff it holds that $\tilde{F}(x^*) \geq \tilde{F}(x)$ for all $x \in X$.

Definition 4.2. A point $x^* \in X$ is said to be a non-dominated solution to Problem (FMP) iff there is no $x \in X$ such that $\tilde{F}(x) \geq \tilde{F}(x^*)$ and $\tilde{F}(x^*) \neq \tilde{F}(x)$.

Definition 4.3. A point $x^* \in X$ is said to be a weak non-dominated solution to Problem (FMP) iff there is no $x \in X$ such that $\tilde{F}(x) > \tilde{F}(x^*)$.

From Definition 3.1, it holds that (1) a feasible solution $x^* \in X$ is an optimal solution to Problem (FMP) if and only if $\tilde{F}(x^*)_{\alpha} \geq_S \tilde{F}(x)_{\alpha}$ for all $x \in X$ and for all $\alpha \in [0, 1]$, (2) $x^* \in X$ is a non-dominated solution to Problem (FMP) if and only if there is no $x \in X$ such that $\tilde{F}(x)_{\alpha} \geq_S \tilde{F}(x^*)_{\alpha}$ holds for all $\alpha \in [0, 1]$ and $\tilde{F}(x)_{\alpha} \neq_S \tilde{F}(x^*)_{\alpha}$ holds for at least one $\alpha \in [0, 1]$, and (3) $x^* \in X$ is a weak non-dominated solution to Problem (FMP) if and only if there is no $x \in X$ such that $\tilde{F}(x)_{\alpha} >_S \tilde{F}(x^*)_{\alpha}$ holds for all $\alpha \in [0, 1]$.

From the above, we define the following optimization problem with set-valued objective function:

$$(\text{SOP})_{\alpha} \quad \begin{cases} \text{maximize} & F_{\alpha}(x) \\ \text{subject to} & x \in X, \end{cases}$$

where $\alpha \in [0, 1]$ is a parameter, and $F_{\alpha} : \mathbb{R}^n \rightsquigarrow \mathbb{R}^{\ell}$ is the set-valued function defined by $F_{\alpha}(x) \equiv \tilde{F}(x)_{\alpha}$. In case that $\ell = 1$, Problem $(\text{SOP})_{\alpha}$ is called an optimization problem with an interval-valued objective function.

It is natural that we define three types of optimal solutions to Problem $(\text{SOP})_{\alpha}$ as follows.

Definition 4.4. A point $x^* \in X$ is said to be an optimal solution to Problem $(\text{SOP})_{\alpha}$ iff it holds that $F_{\alpha}(x^*) \geq_S F_{\alpha}(x)$ for all $x \in X$.

Definition 4.5. A point $x^* \in X$ is said to be a Pareto optimal solution to Problem $(\text{SOP})_{\alpha}$ iff there is no $x \in X$ such that $F(x)_{\alpha} \geq_S F_{\alpha}(x^*)$ and $F_{\alpha}(x) \neq_S F_{\alpha}(x^*)$ hold.

Definition 4.6. A point $x^* \in X$ is said to be a weak Pareto optimal solution to Problem $(\text{SOP})_{\alpha}$ iff there is no $x \in X$ such that $F_{\alpha}(x) >_S F_{\alpha}(x^*)$ holds.

Namely, Problem (FMP) is equivalent to the family of set-valued optimization problems $\{(\text{SOP})_{\alpha}\}_{\alpha \in [0,1]}$, or the set-valued optimization problem with infinite number of set-valued objective functions. Alonzo [1] has investigated optimization problems with a single set-valued function, however, our problem is optimization problems with infinite number of set-valued objective functions and it is not easy to characterize optimal solutions. In order to avoid this, we introduce some conditions under which Problem (FMP) is equivalent to an optimization problems with finite number of set-valued objective functions.

Definition 4.7. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^{\ell})$ be any fuzzy vector-valued function and let $x, y \in \mathbb{R}^n$ be any points. Then $\tilde{F}$ is said to be order preserving on $\mathbb{R}^n$ if $\tilde{F}(x)_1 \geq_S \tilde{F}(y)_1$ and $\tilde{F}(x)_0 \geq_S \tilde{F}(y)_0$ imply $\tilde{F}(x) \geq \tilde{F}(y)$.

Definition 4.8. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^{\ell})$ be any fuzzy vector-valued function and let $x, y \in \mathbb{R}^n$ be any points. Then $\tilde{F}$ is said to be strictly order preserving on $\mathbb{R}^n$ if $\tilde{F}$ is order preserving and $\tilde{F}(x)_1 >_S \tilde{F}(y)_1$ and $\tilde{F}(x)_0 >_S \tilde{F}(y)_0$ imply $\tilde{F}(x) > \tilde{F}(y)$.

For $\tilde{a}, \tilde{b} \in \mathcal{F}_{LR}(\mathbb{R})$, it is well known that $\tilde{a}_0 \geq \tilde{b}_0$ and $\tilde{a}_1 \geq \tilde{b}_1$ imply $\tilde{a} \geq \tilde{b}$ and $\tilde{a}_0 > \tilde{b}_0$ and $\tilde{a}_1 > \tilde{b}_1$ imply $\tilde{a} > \tilde{b}$. Hence we have the following theorem.

Theorem 6. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R})$ be any fuzzy-valued function. Suppose that, for each $x \in X$, $\tilde{F}(x)$ is LR fuzzy number. Then $\tilde{F}$ is strictly order preserving on $\mathbb{R}^n$.

Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}_r(\mathbb{R}^\ell)$ be any fuzzy vector-valued function and we define $\tilde{F}_i : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R})$ by $\tilde{F}_i(x) \equiv (\tilde{F}(x), e^i)$, $i = 1, 2, \dots, \ell$. Then we have the following corollary.

Corollary 4.1. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}_r(\mathbb{R}^\ell)$ be any fuzzy vector-valued function. If $\tilde{F}_i(x)$, $i = 1, 2, \dots, \ell$ are LR fuzzy numbers for each $x \in \mathbb{R}^n$, then $\tilde{F}$ is strictly order preserving on $\mathbb{R}^n$.

In the following, we assume that $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^\ell)$ is order preserving or strictly order preserving on $\mathbb{R}^n$.

Now we define the set-valued function $F : \mathbb{R}^n \rightsquigarrow \mathbb{R}^\ell \times \mathbb{R}^\ell$ by $F(x) \equiv \tilde{F}(x)_1 \times \tilde{F}(x)_0$. We consider the following optimization problem with set-valued objective function:

$$(SOP) \quad \begin{cases} \text{maximize} & F(x) \\ \text{subject to} & x \in X. \end{cases}$$

The following theorems show that Problem (FMP) is equivalent to Problem (SOP) under the assumption that $\tilde{F}$ is order preserving, or strictly order preserving.

Theorem 7. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^\ell)$ be order preserving. Then a point $x^* \in X$ is an optimal solution to Problem (FMP) if and only if $x^* \in X$ is an optimal solution to Problem (SOP).

Proof. Let $x^* \in X$ be any optimal solution to Problem (FMP). Then by definition, it holds $\tilde{F}(x^*)_\alpha \geq_S \tilde{F}(x)_\alpha$ for all $x \in X$ and for all $\alpha \in [0, 1]$. In particular, for $\alpha = 0, 1$, $\tilde{F}(x^*)_0 \geq_S \tilde{F}(x)_0$ and $\tilde{F}(x^*)_1 \geq_S \tilde{F}(x)_1$ hold. Therefore we have $F(x^*) \geq_S F(x)$ for all $x \in X$.

Conversely, let $x^* \in X$ be any optimal solution to Problem (SOP). Then it holds that $F(x^*)_1 \geq_S F(x)_1$ and $F(x^*)_0 \geq_S F(x)_0$ for all $x \in X$. Since $\tilde{F}$ is order preserving, we have $\tilde{F}(x^*)_\alpha \geq_S \tilde{F}(x)_\alpha$ for all $\alpha \in [0, 1]$, which implies that $\tilde{F}(x^*) \geq \tilde{F}(x)$. $\square$

Theorem 8. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^\ell)$ be order preserving. Then a point $x^* \in X$ is a non-dominated solution to Problem (FMP) if and only if $x^* \in X$ is a Pareto optimal solution to Problem (SOP).

Proof. Let $x^* \in X$ be any non-dominated solution to Problem (FMP). On the contrary, suppose that there exists a point $\bar{x} \in X$ such that $F(\bar{x}) \geq_S F(x^*)$ and $F(\bar{x}) \neq_S F(x^*)$ hold. Then by simple calculations, we have $\tilde{F}(\bar{x})_1 \geq_S \tilde{F}(x^*)_1$, $\tilde{F}(\bar{x})_0 \geq_S \tilde{F}(x^*)_0$, and $\tilde{F}(\bar{x})_1 \neq_S \tilde{F}(x^*)_1$ or $\tilde{F}(\bar{x})_0 \neq_S \tilde{F}(x^*)_0$. Since $\tilde{F}$ is order preserving, we have $\tilde{F}(\bar{x})_\alpha \geq_S \tilde{F}(x^*)_\alpha$, $\forall \alpha \in [0, 1]$ and $\tilde{F}(\bar{x})_\alpha \neq_S \tilde{F}(x^*)_\alpha$. Namely we have $\tilde{F}(\bar{x}) \geq \tilde{F}(x^*)$ and $\tilde{F}(\bar{x}) \neq \tilde{F}(x^*)$. This is a contradiction.

Conversely, let $x^* \in X$ be any Pareto optimal solution to Problem (SOP). Suppose that there exists a point $\bar{x} \in X$ such that $\tilde{F}(\bar{x}) \geq \tilde{F}(x^*)$ and $\tilde{F}(\bar{x}) \neq \tilde{F}(x^*)$. Then by definition, we have $\tilde{F}(\bar{x})_\alpha \geq_S \tilde{F}(x^*)_\alpha$ for all $\alpha \in [0, 1]$. Since a point x^* is a Pareto optimal solution to Problem (SOP), it holds that $\tilde{F}(\bar{x})_0 =_S \tilde{F}(x^*)_0$ and $\tilde{F}(\bar{x})_1 =_S \tilde{F}(x^*)_1$. Since $\tilde{F}$ is order preserving, it holds that $\tilde{F}(\bar{x})_\alpha \leq_S \tilde{F}(x^*)_\alpha$ for all $\alpha \in [0, 1]$. Hence we have $\tilde{F}(\bar{x}) = \tilde{F}(x^*)$, which is a contradiction. $\square$

Theorem 9. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R}^\ell)$ be strictly order preserving. Then a point $x^* \in X$ is a weak non-dominated solution to Problem (FMP) if and only if $x^* \in X$ is a weak Pareto optimal solution to Problem (SOP).

Proof. Let $x^* \in X$ be any weak non-dominated solution to Problem (FMP). On the contrary, suppose that there exists a point $\bar{x} \in X$ such that $F(\bar{x}) >_S F(x^*)$ holds. Then by simple calculations, we have $\tilde{F}(\bar{x})_1 >_S \tilde{F}(x^*)_1$ and $\tilde{F}(\bar{x})_0 >_S \tilde{F}(x^*)_0$. Since $\tilde{F}$ is strictly order preserving, we have $\tilde{F}(\bar{x})_\alpha >_S \tilde{F}(x^*)_\alpha$, $\forall \alpha \in [0, 1]$. Namely we have $\tilde{F}(\bar{x}) \succ \tilde{F}(x^*)$. This is a contradiction.

Conversely, let $x^* \in X$ be any weak Pareto optimal solution to Problem (SOP). Suppose that there exists a point $\bar{x} \in X$ such that $\tilde{F}(\bar{x}) \succ \tilde{F}(x^*)$. Then by definition, we have $\tilde{F}(\bar{x})_\alpha >_S \tilde{F}(x^*)_\alpha$ for all $\alpha \in [0, 1]$. In particular,

for $\alpha = 0, 1$, we have $\tilde{F}(\bar{x})_0 >_S \tilde{F}(x^*)_0$ and $\tilde{F}(\bar{x})_1 >_S \tilde{F}(x^*)_1$. Namely we have $F(\bar{x}) >_S F(x^*)$, which is a contradiction. $\square$

Next we consider the case that $\inf \tilde{F}(x)_0$ and $\sup \tilde{F}(x)_0$ are singletons for each $x \in X$. Then Problem (SOP) is equivalent to the following multicriteria optimization problem:

$$(MOP) \quad \begin{cases} \text{maximize} & (\tilde{F}(x)_1, \inf \tilde{F}(x)_0, \sup \tilde{F}(x)_0)^T \\ \text{subject to} & x \in X. \end{cases}$$

Note that, if $\tilde{F}(x)_1 = (1/2)(\inf \tilde{F}(x)_0 + \sup \tilde{F}(x)_0)$ for all $x \in \mathbb{R}^n$ holds, then Problem (MOP) is equivalent to the following problem:

$$(BOP) \quad \begin{cases} \text{maximize} & (\inf \tilde{F}(x)_0, \sup \tilde{F}(x)_0)^T \\ \text{subject to} & x \in X. \end{cases}$$

From the above discussions, we have the following theorems which are well-known results in fuzzy mathematical programming problems with non-interactive fuzzy numbers.

Theorem 10. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R})$ be order preserving. Then a point $x^* \in X$ is an optimal solution to Problem (FMP) if and only if $x^* \in X$ is an optimal solution to Problem (MOP).

Theorem 11. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R})$ be order preserving. Then a point $x^* \in X$ is a non-dominated solution to Problem (FMP) if and only if $x^* \in X$ is a Pareto optimal solution to Problem (MOP).

Theorem 12. Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}(\mathbb{R})$ be strictly order preserving. Then a point $x^* \in X$ is a weak non-dominated solution to Problem (FMP) if and only if $x^* \in X$ is a weak Pareto optimal solution to Problem (MOP).

Next we investigate the properties of optimal value $\tilde{F}(x)$ in (FMP) by using possibility and necessity measures.

Let $\tilde{F} : \mathbb{R}^n \rightarrow \mathcal{F}_r(\mathbb{R}^\ell)$ be any fuzzy vector-valued function and x^* be any optimal solution to Problem (FMP). For each $\alpha \in [0, 1]$, we set $b_\alpha^R \equiv \sup \tilde{F}(x^*)_\alpha$, $b_\alpha^L \equiv \inf \tilde{F}(x^*)_\alpha$. We assume that, for each $x \in X$, $\tilde{F}(x)$ is robust, and $\sup \tilde{F}(x)_\alpha$ is continuous on $[0, 1]$ and strictly decreasing with respect to $\alpha \in [0, 1]$, while $\inf \tilde{F}(x)_\alpha$ are continuous on $[0, 1]$ and strictly increasing with respect to $\alpha \in [0, 1]$.

Then we have $\text{Pos}(\tilde{F}(x^*) \geq b_\alpha^R) = \alpha$, $\text{Nes}(\tilde{F}(x^*) \geq b_\alpha^L) = 1 - \alpha$, and

$$\begin{pmatrix} \text{Pos}(\tilde{F}(x^*) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(x^*) \geq b_\alpha^L) \end{pmatrix} \geq \begin{pmatrix} \text{Pos}(\tilde{F}(x) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(x) \geq b_\alpha^L) \end{pmatrix} \quad \forall x \in X.$$

In fact, we suppose that there exists a point $\bar{x} \in X$ such that $\text{Pos}(\tilde{F}(\bar{x}) \geq b_\alpha^R) \equiv \bar{\alpha} > \text{Pos}(\tilde{F}(x^*) \geq b_\alpha^R) = \alpha$ holds. Then by Corollary 3.1, we have $\sup \tilde{F}(\bar{x})_{\bar{\alpha}} \geq b_\alpha^R$. Since $\sup \tilde{F}(\bar{x})_\alpha > \sup \tilde{F}(\bar{x})_{\bar{\alpha}} \geq b_\alpha^R$, which is a contradiction. On the other hand, we suppose that there exists a point $\bar{x} \in X$ such that $\text{Nes}(\tilde{F}(\bar{x}) \geq b_\alpha^L) \equiv 1 - \bar{\alpha} > 1 - \alpha$ holds. Then by Corollary 3.1, we have $\inf \tilde{F}(\bar{x})_{\bar{\alpha}} \geq b_\alpha^L$ holds. On the other hand, if x^* is a non-dominated solution to Problem (FMP), there is no $\bar{x} \in X$ such that

$$\begin{pmatrix} \text{Pos}(\tilde{F}(\bar{x}) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(\bar{x}) \geq b_\alpha^L) \end{pmatrix} \geq \begin{pmatrix} \text{Pos}(\tilde{F}(x^*) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(x^*) \geq b_\alpha^L) \end{pmatrix} \quad \forall \alpha \in [0, 1].$$

Similarly, if x^* is a weak non-dominated solution to Problem (FMP), there is no $\bar{x} \in X$ such that

$$\begin{pmatrix} \text{Pos}(\tilde{F}(\bar{x}) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(\bar{x}) \geq b_\alpha^L) \end{pmatrix} > \begin{pmatrix} \text{Pos}(\tilde{F}(x^*) \geq b_\alpha^R) \\ \text{Nes}(\tilde{F}(x^*) \geq b_\alpha^L) \end{pmatrix} \quad \forall \alpha \in [0, 1].$$

Let $x^* \in X$ be any feasible solutions to (FMP) and $b^L \equiv \inf \tilde{F}(x^*)_0$, $b^R \equiv \sup \tilde{F}(x^*)_0$. Now we consider the following bi-criteria maximization Problem (PNP):

$$(PNP) \quad \begin{cases} \text{maximize} & (\text{Pos}(\tilde{F}(x) \geq b^R), \text{Nes}(\tilde{F}(x) \geq b^L))^T \\ \text{subject to} & x \in X. \end{cases}$$

Then we have the following theorem:

Theorem 13. *Let $x^* \in X$ be any feasible solution to Problem (PNP). Then x^* is a weak non-dominated solution to Problem (PNP) if and only if x^* is a weak Pareto optimal solution to Problem (FMP).*

Proof. Let x^* be any weak Pareto optimal solution to Problem (PNP). On the contrary, suppose that there exists a feasible solution $\bar{x} \in X$ such that $\tilde{F}(\bar{x}) \succ \tilde{F}(x^*)$ holds. By definition, it holds that $\sup \tilde{F}(\bar{x})_0 > \sup \tilde{F}(x^*)_0 = b^R$ and $\inf \tilde{F}(\bar{x})_0 > \inf \tilde{F}(x^*)_0 = b^L$ hold. By continuity of $\sup \tilde{F}(\bar{x})_\alpha$ with respect to α , we have $\text{Pos}(\tilde{F}(\bar{x}) \geq b^R) > \text{Pos}(\tilde{F}(x^*) \geq b^R)$, which is a contradiction.

Next, let x^* be any weak non-dominated solution to Problem (FMP), and we set $b^R \equiv \sup \tilde{F}(x^*)_0$, $b^L \equiv \inf \tilde{F}(x^*)_0$. On the contrary, we suppose that there exists a feasible solution $\bar{x} \in X$ such that $(\text{Pos}(\tilde{F}(\bar{x}) \geq b^R), \text{Nes}(\tilde{F}(\bar{x}) \geq b^L))^T > (\text{Pos}(\tilde{F}(x^*) \geq b^R), \text{Nes}(\tilde{F}(x^*) \geq b^L))^T$. Hence we have $\text{Pos}(\tilde{F}(\bar{x}) \geq b^R) > \text{Pos}(\tilde{F}(x^*) \geq b^R)$. We set $\bar{\alpha} \equiv \text{Pos}(\tilde{F}(\bar{x}) \geq b^R) > 0$. Then, we have $\sup \tilde{F}(\bar{x})_{\bar{\alpha}} \cap (b^R + \mathbb{R}_+^\ell) \neq \emptyset$, which implies that $\sup \tilde{F}(\bar{x})_{\bar{\alpha}} \geq b^R = \sup \tilde{F}(x^*)_0$. Since $\sup \tilde{F}(x^*)_0 > \sup \tilde{F}(\bar{x})_{\bar{\alpha}}$, we have $\sup \tilde{F}(\bar{x})_0 > \sup \tilde{F}(x^*)_0$. On the other hand, since $\text{Nes}(\tilde{F}(\bar{x}) \geq b^L) = 1$, we have $\inf \tilde{F}(\bar{x})_0 \leq b^L + \mathbb{R}_+^\ell$, which implies that $\inf \tilde{F}(\bar{x})_0 \geq \inf \tilde{F}(x^*)_0$, which is a contradiction. $\square$

Before closing this section, we consider the following fuzzy mathematical programming problem:

Example 4.1.

$$(FMP) \quad \begin{cases} \text{maximize} & \langle \widetilde{(2, 3)}, x \rangle \\ \text{subject to} & x_1 + 3x_2 \leq 9, \\ & 3x_1 + 5x_2 \leq 18, \\ & 5x_1 + 7x_2 \leq 28, \\ & x_1 \geq 0, x_2 \geq 0. \end{cases}$$

where $\widetilde{(2, 3)} \in \mathcal{F}(\mathbb{R}^2)$ is the fuzzy vector defined by

$$\mu_{\widetilde{(2, 3)}}(t_1, t_2) \equiv \max\{1 - (t_1 - 2)^2 - (t_2 - 3)^2, 0\}.$$

We set $X \equiv \{x \in \mathbb{R}^2 \mid x_1 + 3x_2 \leq 9, 3x_1 + 5x_2 \leq 18, 5x_1 + 7x_2 \leq 28, x_1 \geq 0, x_2 \geq 0\}$. Let $x \in X$ be any point in X . Then for each $\alpha \in [0, 1]$, we have

$$\langle \widetilde{(2, 3)}, x \rangle_\alpha = \left[2x_1 + 3x_2 - \sqrt{(1 - \alpha)(x_1^2 + x_2^2)}, 2x_1 + 3x_2 + \sqrt{(1 - \alpha)(x_1^2 + x_2^2)} \right].$$

Moreover, it is easy to see that $\langle \widetilde{(2, 3)}, x \rangle$ is strictly order preserving on $\mathbb{R}_+^n$. Therefore, in order to find all non-dominated or weak non-dominated solutions to Problem (FMP), it suffices to solve the following the bi-criteria nonlinear optimization problem:

$$(MOP) \quad \begin{cases} \text{maximize} & \left(2x_1 + 3x_2 - \sqrt{x_1^2 + x_2^2}, 2x_1 + 3x_2 + \sqrt{x_1^2 + x_2^2} \right)^T \\ \text{subject to} & x \in X. \end{cases}$$

Problem (MOP) is called difference convex optimization problem which is characterized by the property whose objective function is expressed by difference of convex functions.

Next, we consider the case that fuzzy vector $\widetilde{(2, 3)}$ is rectangle and which is given by

$$\mu_{\widetilde{(2,3)}}(t_1, t_2) \equiv \max\{\min\{1 - |t_1 - 2|, 1 - |t_2 - 3|\}, 0\}.$$

Let $x \in X$ be any point in X . Then for each $\alpha \in [0, 1]$, we have

$$\langle \widetilde{(2, 3)}, x \rangle_\alpha = [2x_1 + 3x_2 - (1 - \alpha)(x_1 + x_2), 2x_1 + 3x_2 + (1 - \alpha)(x_1 + x_2)].$$

It is easy to see that $\langle \widetilde{(2, 3)}, x \rangle$ is strictly order preserving on $\mathbb{R}_+^n$. Therefore, in order to find all non-dominated or weak non-dominated solutions to Problem (FMP), it suffices to solve the following parametric linear programming problem [13,14]:

$$(LP_\lambda) \quad \begin{cases} \text{maximize} & (2 + \lambda)x_1 + (3 + \lambda)x_2 \\ \text{subject to} & x \in X, \end{cases}$$

where $\lambda \in [-1, 1]$ is a parameter. Let $\tilde{c}_1, \tilde{c}_2 \in \mathcal{F}(\mathbb{R})$ be any symmetric L fuzzy numbers given by $\tilde{c}_1 \equiv (1, 1)_L$, $\tilde{c}_2 \equiv (12, 3)_L$ and let $d^1 \equiv (1, -1)^T$, $d^2 \equiv (2, 3)^T \in \mathbb{R}^2$ be any vectors. Then we define the fuzzy vector $\tilde{c} \equiv (\tilde{c}_1, \tilde{c}_2) \in \mathcal{F}(\mathbb{R}^2)$ by

$$\begin{aligned} \mu_{\tilde{c}}(r) &\equiv \min\{\mu_{\tilde{c}_1}(\langle r, d^1 \rangle), \mu_{\tilde{c}_2}(\langle r, d^2 \rangle)\} \\ &= \min\{\mu_{\tilde{c}_1}(r_1 - r_2), \mu_{\tilde{c}_2}(2r_1 + 3r_2)\} \quad \forall r \in \mathbb{R}^2. \end{aligned}$$

The fuzzy vector $\tilde{c}$ is called oblique fuzzy vector [7].

Then we consider the following problem:

$$(FMP2) \quad \begin{cases} \text{maximize} & \langle \widetilde{(3, 2)}, x \rangle \\ \text{subject to} & x \in X. \end{cases}$$

Then for each $\alpha \in [0, 1]$, we have

$$\begin{aligned} \widetilde{(3, 2)}_\alpha &\equiv \text{conv}(\{(3, 2 + (1 - \alpha))^T, (3 + \frac{6}{5}(1 - \alpha), 2 + \frac{1}{5}(1 - \alpha))^T, (3, 2 - (1 - \alpha))^T \\ &\quad (3 - \frac{6}{5}(1 - \alpha), 2 - \frac{1}{5}(1 - \alpha))^T\}), \end{aligned}$$

where the $\text{conv}(A)$ denotes the convex hull of the set A . We set $x^1 \equiv (3, 3)^T$, $x^2 \equiv (\frac{21}{5}, \frac{11}{5})^T$, $S_1 \equiv \widetilde{(3, 2)}_0 + \mathbb{R}_-^2$ and $S_2 \equiv \widetilde{(3, 2)}_0 + \mathbb{R}_-^2$. Let $N(x^1; S_1)$ be the normal cone to the set S_1 at x^1 and let $\alpha \in [0, 1]$ be any number. Then, by simple calculation, we have

$$\langle \widetilde{(3, 2)}, x \rangle_\alpha = [3x_1 + 2x_2 - (1 - \alpha)x_2, 3x_1 + 2x_2 + (1 - \alpha)x_2] \quad \forall x \in N(x^1; S_1).$$

While, let $N(x^2; S_2)$ be the normal cone to the set S_2 at x^2 and let $\alpha \in [0, 1]$ be any real number. Then, for all $x \in N(x^2; S_2)$, we have

$$\langle \widetilde{(3, 2)}, x \rangle_\alpha = [3x_1 + 2x_2 - (1 - \alpha)(\frac{6}{5}x_1 + \frac{1}{5}x_2), 3x_1 + 2x_2 + (1 - \alpha)(\frac{6}{5}x_1 + \frac{1}{5}x_2)].$$

Moreover, it is easy to prove that $\langle \widetilde{(3, 2)}, x \rangle$ is strictly order preserving on $\mathbb{R}_+^2$.

Noting that $N(x^1; S_1) \cup N(x^2; S_2) = \mathbb{R}_+^2$, in order to find all non-dominated or weak non-dominated solutions to Problem (FMP2), it suffices to solve the following piecewise linear bi-criteria optimization problem:

$$(MOP2) \quad \begin{cases} \text{maximize} & (\inf \langle \widetilde{(3, 2)}, x \rangle_0, \sup \langle \widetilde{(3, 2)}, x \rangle_0)^T \\ \text{subject to} & x \in X. \end{cases}$$

In [7], Inuiguchi et al. proposed another methods which transformed fuzzy optimization problems with oblique fuzzy vectors to linear programming problems by using necessity measures.

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